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Physics-Informed Bayesian Event Attribution From Sparse PMUs

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ABSTRACT Eight PMUs observe only part of an IEEE 39-bus system, so a disturbance can be detected without uniquely identifying its cause. We evaluate a recovery-aware physical hypothesis bank for continuous attribution of load changes and line openings in a fixed simulated catalog. Switched differential-algebraic and oriented PMU-terminal models generate candidate observation paths; a local truncated-Gaussian integral handles unknown load severity, and the finite history bank retains predicted physical and instrumental response after OFF. The full bank remains approximate through plug-in context, shared regime covariances, screening and pruning. Across 18 six-intervention schedules, three network conditions and two noise scales, it assigns the correct family and source before OFF in 644 of 648 event realizations, with 72 unmatched alarms. Two causal ExtraTrees references reach 330/648 with 29 alarms and 352/648 with 16 alarms; both were weak on load attribution during development, so the comparison does not establish general superiority over machine learning. A posthoc audit finds that 0/644 available physical-bank restorations and 0/643 nominal restorations occur more than one report early, versus 52/330 and 26/352 for the learned methods; many learned events have no restoration value. These dependent replays do not isolate the effect of memory. The results support a coverage–alarm tradeoff for this model family and catalog, not calibrated probabilities, field validation or real-time operation.

INDEX TERMS event detection, model uncertainty, phasor measurement units, physics-informed inference, sequential diagnosis, source localization.

I. INTRODUCTION

EIGHT PMUs observe only a small part of a 39-bus network, yet an operator needs to know whether a disturbance occurred, which physical mechanism caused it, and where it began. An unmeasured bus can change several remote PMU signals, while different sources can produce similar measured responses. The problem is therefore both electrical and statistical: reconstruct the response allowed by the network, then compare its possible causes under uncertain state, severity and noise.

Our starting point is a feature-based hackathon system. Voltage, current, frequency and spatial/temporal descriptors supported an ExtraTrees detector, classifier and localizer. We retain that route and its results. The complementary route asks an explicit counterfactual question: if this source had changed at a candidate past time, what would all eight PMUs have measured up to now? Physical equations provide the candidate path; conditional Bayesian evidence accounts for unknown severity and competing histories. After restoration, those histories must still explain the remaining machine motion.

Physical restoration does not erase machine motion. A flat reference can mislabel its remaining oscillation as a second event; suppressing all alarms during recovery can hide an actual second event. A corrected voltage offset also need not correct the response to a later input.

DAE estimators recover unmeasured electrical states [1]; outage detectors compare physical hypotheses [2], [3]; and simulated trajectories support sparse-PMU fault diagnosis [4]. These ingredients are established. Our question is whether a continuing recovery should explain a new PMU change or yield to a later intervention hypothesis.

We formulate a causal attribution procedure that keeps the estimated physical response after an intervention ends and tests subsequent changes against that continuing response. The paper makes three bounded contributions:

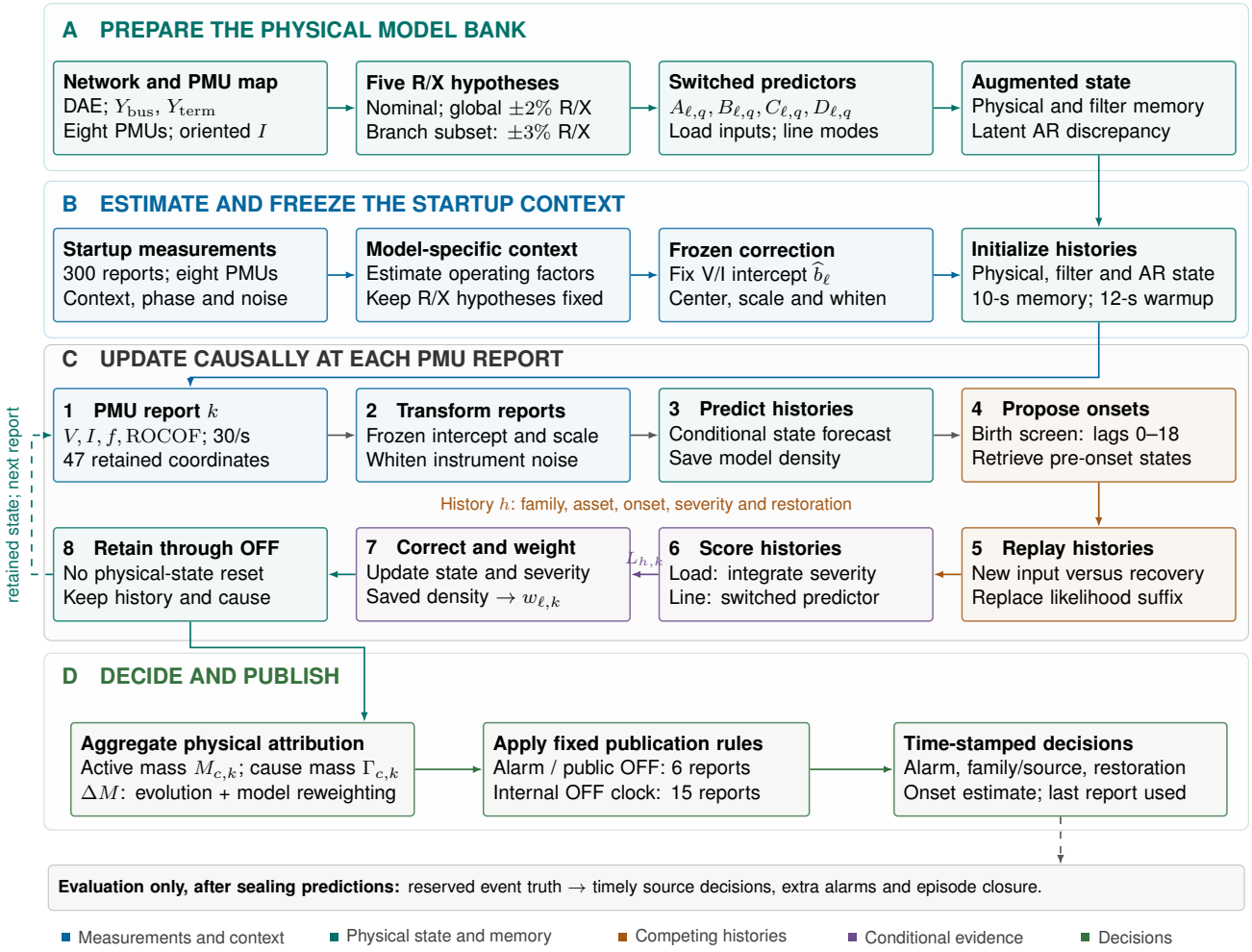


FIGURE 1. Implemented physical inference in four stages. (A) Construct switched network and PMU predictors. (B) Estimate the startup context and fix its observation correction. (C) Compare new intervention histories with retained recovery at each report, using stored pre-onset states. (D) Aggregate attribution and publish qualified alarm, source and restoration decisions. The public six-report rules and the internal 15-report OFF clock are distinct; their support conditions are specified in Section IV. Dashed feedback retains state for the next report. Reserved truth enters only the later evaluation.

1) It specifies an executable physical measurement contract in which a switched DAE and oriented terminal map generate the candidate PMU paths.
2) It evaluates a continuous history bank that compares new load or line interventions with retained recovery, integrates local load severity, and reports active mass M separately from originating-cause mass Γ. These are summaries of an approximate, screened hypothesis bank.
3) It measures timely family/source attribution, unmatched alarms and restoration behavior on a prespecified catalog, and states exactly how that evidence differs from two frozen ExtraTrees procedures and historical window-based results.

The complete physical bank attributes 322/324 load realizations before OFF versus 15/324 for the causal ExtraTrees reference. Under out-of-bank R/X it raises 68 versus 10 unmatched alarms. Thus the supported gain is timely load attribution with a larger alarm burden. A separate posthoc component study tests amplitude-evidence scoring and physical-state carry at inferred closure. It is not an independent confirmation and cannot establish a benefit until all four arms and their intervals are reviewed.

The test uses nine candidate load buses and nine lines in a fixed IEEE-39 simulation. It does not establish an all-family hackathon score, calibrated event probabilities or real-time deployment.

II. RELATED WORK

DAE and unknown-input estimators recover hidden states or inputs [1], [5], [6]; functional observability and diagnosability ask which states or sources can be distinguished [7]–[10]. Physical outage inference already uses network hypotheses, uncertain states and sequential tests [2], [3], [11], [12]. Sparse input recovery estimates forcing locations and unknown initial states in linear dynamics [13]. Sensor placement limits the information available at remote PMUs [14]–[16].

Warner *et al.* compare simulated voltage/current paths

across an unknown-onset window for short-circuit line identification [4]. Yang, Chen and Zhai combine partial-PMU algebraic measurements with particle-filter estimates of generator dynamic states, then monitor outage signatures with an EWMA on the IEEE-39 system [17]. This is a close physical precedent for line outage detection; the present evaluation instead attributes load and line interventions through OFF and successive recovery. Neither study is run as a same-data benchmark here. Multiple-model filters and adaptive history banks already retain conditional modes [18], [19]; sequential diagnosis has established error/delay theory under specified observation laws [20]. Our screened, pruned bank inherits none of those general optimality or delay guarantees.

Learned PMU classifiers and physical uncertainty models provide other routes [21]–[27]. For example, Li *et al.* embed disturbance-type/topology and localization-cost constraints in a spatiotemporal graph-convolutional model [22]. Our physical route instead computes the observation likelihood of candidate interventions; its graph is the admittance network and terminal map, not a learned message-passing layer. The two frozen ExtraTrees implementations test supervised references, not all learned diagnosis. The new contribution claimed here is the specified sparse-PMU procedure and its evidence: new interventions and retained recovery histories compete for the same causal reports, with component experiments explaining the role and limits of physical representation. We do not claim that combining physics, Bayes and a tensor is itself new. Table 1 distinguishes the neighboring tasks and the comparators actually executed here.

III. PROBLEM AND MEASUREMENT MODEL

At report k, inference uses measurements through $t_k = k/30$ to decide whether a physical intervention occurred. Detection concerns its presence; classification selects a physical operator; localization selects a bus or branch. These outputs can become reliable at different times. We record the estimated physical onset $\widehat{\tau}$ separately from the alarm and diagnosis emission times.

Figure 2 introduces the broader measurement phenomena using the known RAW0001 reference record. Faults, line changes, generation changes and load changes alter the physical network; missing or corrupted reports alter observation integrity. The present causal test addresses load changes and line openings with restoration. The other panels provide context, not additional validated event classes.

The network has 39 buses, but the prescribed PMU set is

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}. \tag{1}$$

Voltage is measured at each PMU bus. Current is measured at a specified branch terminal, which can differ from that bus. The causal implementation uses 47 coordinates: eight voltage magnitudes, eight voltage angles, eight current magnitudes, seven current angles, eight absolute frequencies and eight ROCOFs, at 30 reports/s. BUS39 current angle is excluded because its interpretation remains partially validated. The BUS5 current terminal retains the provisional historical 5–6 assignment. These are declared measurement assumptions, not identities inferred from classification accuracy.

The exported magnitudes are in V and A, angles in degrees, frequency in Hz and ROCOF in Hz/s. Internal centered coordinates do not replace the absolute frequency contract. These are reporting-rate phasor measurements; waveform harmonics or THD are not measured by these channels.

The physical model is

$$\dot{x} = f_q(x, z, u; \theta), \qquad 0 = g_q(x, z, u; \theta), \tag{2}$$

$$y_k = \mathcal{M}_k(x_{0:k}, z_{0:k}, q_{0:k}; \theta_M) + b + \varepsilon_k. \tag{3}$$

Here x contains machine and control dynamics, z the algebraic network variables, q the topology/intervention mode, and θ_M the measurement convention. The causal measurement map uses current algebraic variables for voltage and terminal current and retained phase/instrument history for frequency and ROCOF. The history notation makes that memory explicit; the instantaneous electrical map used in the Schur reduction below is its voltage/current component. A machine's rotor speed is not substituted for bus frequency.

An intervention history is $h = (e, c, \tau, a, \kappa)$: family, source, onset, severity and restoration time, with $\kappa = \infty$ for a sustained event. Its duration is $\kappa - \tau$ when restoration occurs. A load changes its local P/Q setpoints; a line opening switches its branch primitive. For the supported impulse-free switches,

$$x(\tau^+) = x(\tau^-), \qquad g_{q^+}(x(\tau^-), z(\tau^+), u; \theta) = 0. \tag{4}$$

The algebraic voltage/current response may jump even though dynamic states remain continuous. Physical restoration reverses the intervention operator. It does not return the machines to their pre-event state, so electrical recovery can continue after OFF.

The generator stores event schedules and true states separately from the PMU stream. Inference receives the nominal network, permitted calibration and reports through the current time. It receives no source, switch time, true R/X realization or solver state. Initial operating context and noise scale are estimated from 300 startup reports; monitoring begins after the inherited 12-s warmup. The startup contract assumes no intervention in that prefix.

IV. RECOVERY-AWARE PHYSICAL ATTRIBUTION

Figure 1 gives the report-level logic. A history is a proposed cause and trajectory, not just a bus label. The inference bank predicts the next report under no new event, a new load or line action, or retained recovery. Its switched predictors come from the network DAE and oriented PMU terminal map. Statistical filtering handles uncertain states and residual error. The nonlinear simulator generates the observations; inference uses local predictors, not a new full simulation for every hypothesis and report.

TABLE 1. Closest inference contracts. Chen and the two fixed ExtraTrees adapters are the executed comparators. An untested capability is not an algorithmic impossibility.

Route	Mechanism	Evaluated target	Role in this study
Chen / Rovatsos	Network likelihoods and sequential tests	Line outages; Rovatsos includes transients	Chen on persistent-line first decisions
Warner high-fidelity	Simulated V/I paths and onset search	Short-circuit line candidates	Closest trajectory precedent; no common-data run
Yang–Chen–Zhai (2021)	Particle filter for generator states; EWMA outage monitoring	Line-outage detection with partial PMUs on IEEE-39	Closely related line detector; no continuous load/restoration comparison
Adaptive Matrix CuSum	Sequential diagnosis with error/delay analysis	One unknown distributional change	Statistical precedent; no inherited guarantee
DAE / stochastic-state estimation	Physical state and uncertainty models	States, unknown inputs or parameters	Components and adjacent tasks
Learned PMU diagnosis	Features or constrained representations	Event families and locations	Two frozen ExtraTrees references
Physical history bank here	Estimated-past-state paths and recovery	Successive catalog load/line events	Timely attribution, extras and restoration

TABLE 2. Scope of the executed causal component. The source is an intervention asset; it is not necessarily a measured PMU bus.

Case	Physical operator	Executed claim
Normal	No intervention	Monitor after startup
Load	Local P/Q factor; optional restoration	Nine candidate buses
Line	Branch open; optional reclose	Nine candidate branches
Fault/generation	Different shunt/input/trip operators	Outside this causal test
Missing/bad data	Measurement-integrity mechanism	Outside causal test
Unknown	Unresolved diagnosis	Not a new physical family

A. CONSTRUCT THE PREDICTOR FROM THE NETWORK

Kirchhoff constraints and complex power connect sources to remote PMUs:

$$I_{\mathrm{bus}} = Y_{\mathrm{bus}} V, \quad I_{term,r} = Y_{term,r} V, \quad S_i = V_i I^*_{\mathrm{bus},i}. \tag{5}$$

Terminal rows retain orientation, line charging and transformer ratios. A measured terminal current is not a nodal injection. A line opening changes the branch stamp and therefore the algebraic voltages and measured currents. A terminal map changes directly only if the switched branch is itself measured; none of the nine tested line candidates is a measured PMU terminal. The load operator scales both P_{set} and Q_{set} by $1 + a$ with the fixed constant-impedance ZIP fractions, while the line operator switches the selected branch between active and open modes.

The executed plant has 114 differential states, 78 algebraic variables and 38 load-input coordinates. Its ten Sauer–Pai machines use AVR Type I with quadratic ceiling and TGOV1 governors, except the bus-39 machine, which has no external governor or excitation-control inputs. All 19 load models use constant-impedance ZIP fractions. With injection sign convention,

$$P_i = (1+a_i) P_i^0 \left|V_i / V_{set,i}\right|^2, \qquad Q_i = (1+a_i) Q_i^0 \left|V_i / V_{set,i}\right|^2. \tag{6}$$

Here $P_i^0 < 0$ represents consumption; reactive injection can be positive for a capacitive load. The archived equilibrium and voltage setpoints, including those at buses 31 and 39, define the executed operating point.

On a regular index-one DAE chart with a fixed angle reference,

$$\begin{aligned} \delta z &= -g_z^{-1}(g_x \delta x + g_u \delta u), \\ \delta \dot{x} &= A_q^{red} \delta x + B_q^{red} \delta u, \\ A_q^{red} &= f_x - f_z g_z^{-1} g_x, \quad B_q^{red} = f_u - f_z g_z^{-1} g_u. \end{aligned} \tag{7}$$

For the instantaneous electrical observation map $H_q(x, z, u)$, the corresponding output operators are

$$C_q^{red} = H_x - H_z g_z^{-1} g_x, \qquad D_q^{red} = H_u - H_z g_z^{-1} g_u. \tag{8}$$

They propagate a perturbation at an unmeasured source to the selected voltage and oriented-current channels. Network connectivity enters through the branch stamps in g_q and, where applicable, H_q; eliminating unmeasured algebraic variables generally produces dense reduced operators. The graph therefore constrains the prediction without a learned graph embedding. A changed terminal operator would also contribute directly through H_q; the tested line modes keep the same terminal map. The implementation discretizes the admitted switch modes and adds causal phase-history operators for frequency and ROCOF. The generator estimates phase derivatives with a robust, magnitude-weighted quadratic fit over 21 reports. The predictor approximates that operation with a fixed unweighted quadratic fit over the same history; these operators need not agree when weights change. Inertia, controls and load law remain those of the fixed plant.

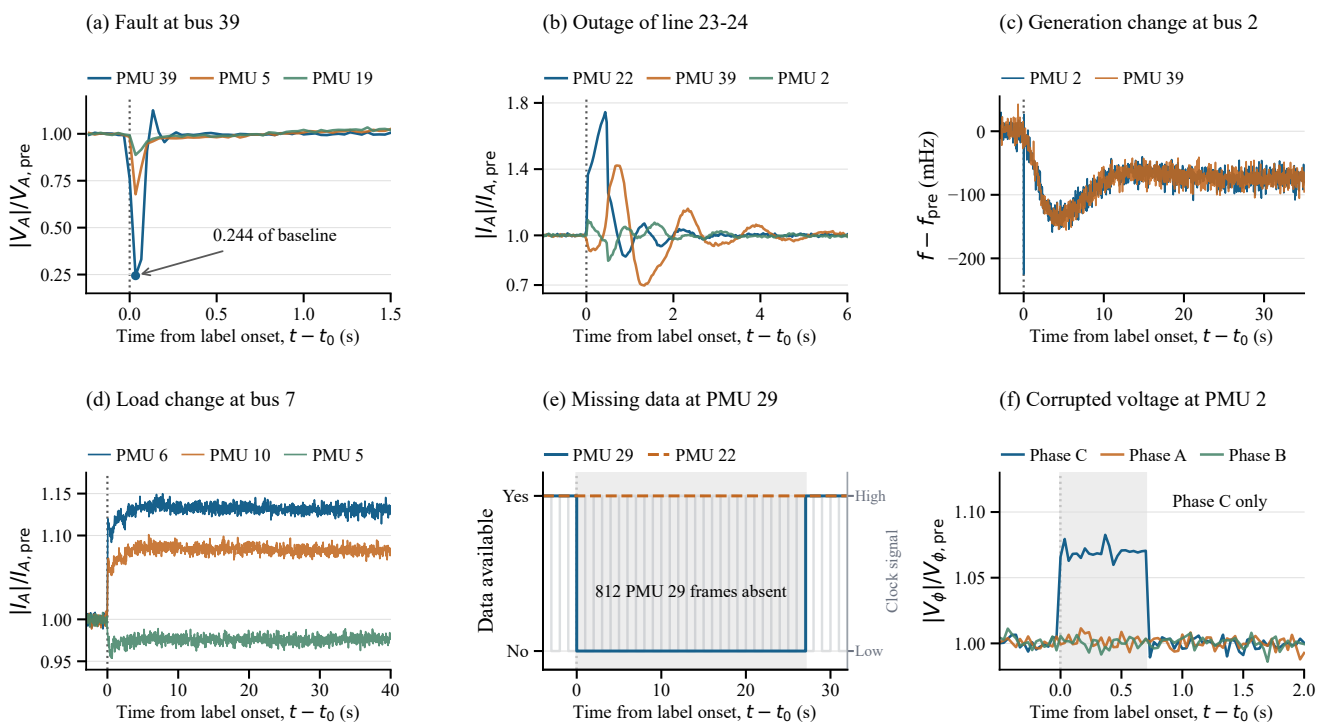


FIGURE 2. Representative signatures from the archived RAW0001 reference simulation: fault, line outage, generation change, load change, missing PMU data and corrupted voltage. Alignment uses retrospective label onset for illustration; inference receives no such alignment. The gray clock in (e) is an illustrative reference, not another measured channel. Only load and line mechanisms enter the causal evaluation below. RAW0001 is known retrospective evidence, not independent field validation.

After augmenting instrumental memory, a conditional predictor has the local form

$$\begin{aligned} \xi_{k+1} &= A_{\ell,q}\xi_k + b_{\ell,q} + B_{\ell,q}a + w_k, \\ r_k &= C_{\ell,q}\xi_k + d_{\ell,q} + D_{\ell,q}a + v_k. \end{aligned} \tag{9}$$

Here r_k contains reference-corrected, scaled and instrument-whitened PMU coordinates, ℓ is a fixed network hypothesis, and ξ_k contains physical deviation states, phase history and predicted measurement-filter memory. Observed colored-instrument filters whiten measurements; their corresponding predictor memory is deterministic. Additional latent AR states represent persistent discrepancy, with preset quiet/load/line/recovery covariance and drift regimes. This statistical discrepancy model is distinct from the plant equations. v_k denotes the remaining innovation. A load has scalar severity within its specified P/Q direction. A line uses a switched operator rather than a fictitious load-amplitude fit. Equation (9) states the generic local model; the implemented update combines propagation and observation into an operator on the previous state. The supplement gives that indexing and the associated transition gain explicitly. States from different equilibria are not interpreted as an absolute rotor-state mixture.

B. ESTIMATE CONTEXT FROM THE STARTUP PREFIX

Each network hypothesis uses the same 300 startup reports to estimate operating factors, reference phase and noise scale. Voltage/current coordinates receive a constant discrepancy correction,

$$\widehat{b}_\ell = \bar{y}_{0:299} - y_\ell^{ref}, \qquad b_{\ell,k} = \widehat{b}_\ell \quad (k \geq 300). \tag{10}$$

Angular differences respect wrapping and the common reference. Frequency and ROCOF use empirical prefix centers. The correction remains fixed so it cannot track a later load step and subtract its signature. This estimates an observation discrepancy, not the true impedances. Prefix estimates enter by substitution rather than through a full joint parameter posterior.

C. COMPARE UNKNOWN-ONSET HISTORIES

The method retains ten seconds of past conditional states. Its frozen birth screen proposes selected onsets at 19 candidate lags, 0–18, within a 20-report candidate window. It does not enumerate every onset in the 300-report history. Each history specifies family, asset, onset, restoration mode and severity. It starts from the stored conditional state before its proposed onset. That state already contains any ongoing recovery; an alarm does not reset the predictor to equilibrium.

For a retained load history, conditional Gaussian filtering yields a residual e, severity response s and covariance S. Unknown severity has a truncated-Gaussian distribution inherited from previous reports:

$$L_{h,k} = \int_l^u \mathcal{N}(e; sa, S)\, \mathrm{TN}_{[l,u]}(a; \mu, v)\, da. \tag{11}$$

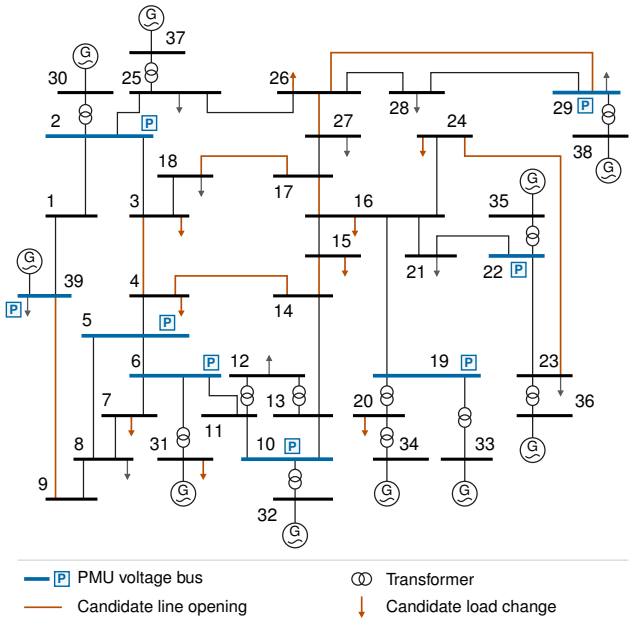


FIGURE 3. **IEEE 39-bus single-line diagram: 46 branches, 12 transformers and 10 generators. Blue bars and P marks locate the eight voltage PMUs. Orange branches and load arrows identify the nine line and nine load candidates; gray arrows denote other loads. Current terminals follow the separate measurement contract. All routes are orthogonal.**

The supplement derives the closed form by completing the square and works through a numerical example. It also specifies the conditional state representation $m_h + d_h a$ and its update. The local amplitude integral is exact for the stated conditional Gaussian model. The complete bank remains approximate: regime covariances are shared from nominal-operator Riccati equations, context estimates enter by substitution, and histories are screened and pruned. At episode closure the augmented state mixture mean is retained and omitted between-history spread is recorded. A fully Bayesian posterior over states, context and all histories is not claimed.

The rolling screen proposes histories; it is not another independent observation. Replaying a suffix replaces its original no-new-event suffix likelihood. The sequential update therefore counts each report once, rather than repeatedly accumulating overlapping ten-second windows.

Within each fixed network hypothesis ℓ, let $\mathcal{H}_{\ell,k}$ denote its screened non-null histories and $\mathcal{A}_{\ell,k} = \mathcal{H}_{\ell,k} \cup \{0\}$ include the null history 0 (no intervention). For each currently admitted candidate h, let $\bar{\pi}_{\ell,h,k}$ denote its predicted mass before scoring the current report. For an existing history this prediction applies the implementation's OFF hazards to survivors and children, generated on demand. For a proposed birth at report τ, it includes the null-parent mass, the frozen birth prior, and the replay likelihood ratio $\exp(\ell^{new}_{\tau:k-1} - \ell^{old}_{\tau:k-1})$ with the corresponding transition and prior terms. Thus the proposal replaces the old no-event suffix before the current report is scored; the reports in the replayed suffix are not counted twice. The supplement gives the corresponding score identity.

The report update and within-hypothesis normalization are

$$\widetilde{\pi}_{\ell,h,k} = L_{\ell,h,k}\bar{\pi}_{\ell,h,k}, \qquad \widetilde{\pi}_{\ell,0,k} = L_{\ell,0,k}\bar{\pi}_{\ell,0,k},$$
$$\pi_{\ell,h,k} = \frac{\widetilde{\pi}_{\ell,h,k}}{\sum_{j\in\mathcal{A}_{\ell,k}} \widetilde{\pi}_{\ell,j,k}}. \tag{12}$$

This is a normalization scheme for the currently admitted candidates, not a full Markov transition over all possible histories. After each update, the screen and beam prune hypotheses and the retained masses are renormalized jointly with the null. The outer weights $w_{\ell,k}$ are updated from network-level predictive densities and combine these conditional masses in M and Γ; they are not part of the within-bank normalizer. The resulting masses do not define a calibrated posterior.

D. REPRESENT FIXED NETWORK UNCERTAINTY

The bank contains nominal, global $\pm 2\%$, and selected-branch $\pm 3\%$ paired R/X hypotheses. Parameters stay constant within a hypothesis. Its outer weight uses a predictive density formed before the current-report state update and proposal selection:

$$w_{\ell,k} = \frac{w_{\ell,k-1} p_\ell(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_\ell)}{\sum_j w_{j,k-1} p_j(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_j)}. \tag{13}$$

Startup-fitting reports are not scored again as independent model-selection evidence. The mixture covers a finite set of network alternatives rather than identifying a continuous impedance posterior.

E. KEEP RECOVERY AS A COMPETING PHYSICAL EXPLANATION

Restoring an input or topology at κ does not restore the pre-event state: $x(\kappa^+) = x(\kappa^-)$. On a regular local post-restoration chart, the difference from a same-context no-new-intervention trajectory obeys

$$\delta x_h(t) \simeq \Phi_0(t,\kappa)\delta x_h(\kappa),$$
$$\delta y^{net}_h(t) \simeq C_0(t)\Phi_0(t,\kappa)\delta x_h(\kappa), \quad t > \kappa. \tag{14}$$

Here y^{net} denotes voltage/current coordinates before the instrument map. Instrumental memory is retained as well. This local interpretation of known dynamics explains why the reference after OFF is a trajectory, not a zero residual; it is not a new propagation theorem.

For conditional history mass $\pi_{\ell,h,k}$, define active and originating-cause masses for an admissible family/source cell c:

$$M_{c,k} = \sum_\ell w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}\, \mathbf{1}\{h \text{ active at } k\},$$
$$\Gamma_{c,k} = \sum_\ell w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}. \tag{15}$$

A retained history can stop being active while its cause still explains recovery. Later interventions compete with those histories. The implementation stores 18 admissible cells, not every family at every bus. After confirmed episode closure, explicit histories are compressed into the retained state mean.

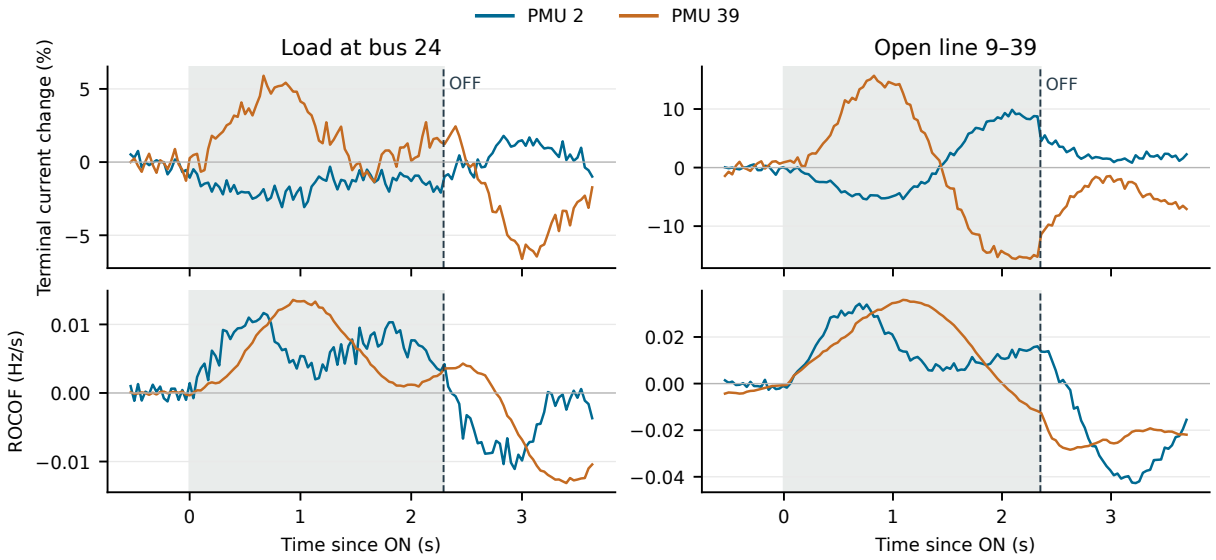


FIGURE 4. **Two interventions from one continuous development record. Current change and ROCOF at PMUs 2 and 39 remain dynamic after physical OFF (dashed line). The gray interval is the true active period, shown only for explanation; inference uses all eight PMUs without event timing. These two PMUs illustrate the traces and were fixed for both panels.**

Cause masses describe the current bank, not a permanent ledger of all past interventions.

Writing the conditional active mass as $M_{\ell,c,k}$ yields

$$\Delta M_{c,k} = \sum_{\ell} w_{\ell,k}(M_{\ell,c,k} - M_{\ell,c,k-1}) + \sum_{\ell}(w_{\ell,k} - w_{\ell,k-1})M_{\ell,c,k-1}. \quad (16)$$

The terms distinguish within-model evolution from model reweighting. This accounting identity aids interpretation, without constituting a new tensor factorization. The masses are approximate attribution weights; probability calibration and globally optimal filtering are not established.

F. EMIT SEPARATE DETECTION, DIAGNOSIS AND RESTORATION MESSAGES

The internal observer proposes an alarm after the six-report persistence rule above the 0.99 active-mass threshold. Public alarm publication additionally requires an internal pending episode and six reports of qualified active support: the frozen source screen qualifies load cells, while admitted line cells qualify through their switched operator. Public OFF requires six reports above 0.99 for the diagnosed cell's joint cause-and-restoration mass. The observer also has a separate, more conservative 15-report internal OFF clock. Diagnosis selects the family with greatest summed cause mass and then the highest-mass source within that family. It is emitted after six reports of stable cause support or at the 15-report post-alarm deadline. The supplement defines the source screen, early line rule and episode replacement, including how internal messages are propagated publicly. Every message records emission time and the last report used. A diagnosis can be correct eventually but too late to precede physical OFF.

The nominal comparator uses the same nominal conditional model with its own fixed publication path. Comparing nominal and mixture evaluates two complete configured procedures; it does not isolate the outer network bank from publication decisions. The admission-rule ablation tests a different publication decision; it is not a pure recovery-memory ablation.

The supplement records the frozen birth/restoration hazards, severity prior and integration intervals, restoration-child rules and beam limits, together with the execution sequence and illustrative calculations. These details preserve the archived configuration; the compact algorithm below states the report order and information boundary.

G. PER-REPORT ALGORITHM AND COMPUTATIONAL BOUNDARY

Algorithm 1: one causal update. Input: nominal model, fixed method configuration and PMU reports. Initialize context, reference, noise and conditional states from the 300 startup reports. Event truth is never an input.

1) Transform report k; predict it from each retained physical or recovery history. Save each network model's predictive density before screening or correcting with this report.
2) Screen 19 candidate onset lags (0–18) within the 20-report window. Start each proposed history from its stored pre-onset state and replay its suffix, replacing the old suffix likelihood once.
3) Score histories, integrating load severity or using the switched line operator. Then correct state and severity, update network weights, and prune by the fixed rules. Retain state and cause through restoration.
4) Aggregate active mass M, cause mass Γ and ΔM.

Apply the frozen alarm, source and OFF rules; record emission time, estimated onset and last report used.

The nonlinear simulator constructs observations; this algorithm propagates local switched predictors. The input grid is 30 reports/s. Causality concerns report availability; acquisition throughput, computation time and the electrical history needed for a decision are separate quantities. Wall-clock real-time operation has not been established.

V. WHAT PHYSICAL STRUCTURE AND LEARNED FEATURES RECOVER

The continuous estimator builds on three distinct questions: what can be reconstructed at unmeasured buses, what response distinguishes candidate sources, and how evidence should be assigned when several explanations fit. The following component experiments answer parts of these questions. They use different cohorts and are not ablations of the final continuous estimator. Figure 5 places their measured effects side by side without pooling their scores.

A. RECOVERING UNMEASURED VOLTAGES IS NOT YET EVENT ATTRIBUTION

In 100 nominal test trajectories, the mean hidden-voltage relative error falls from 0.3402% for the equilibrium reference to 0.00990% for snapshot weighted least squares and 0.00963% for a causal Kalman reconstructor. Most of this improvement comes from conditioning the physical network on the measured phasors; temporal filtering adds a smaller improvement. This supports reconstruction under that model and sensor placement, not unique recovery of every source. The independently evaluated Kalman reconstructor is not silently substituted for the conditional observer in Fig. 1.

B. PHYSICAL FEATURES REMAIN A USEFUL LEARNED ROUTE

The historical feature study raises physical-source Top-1 from 49.6% with 136 features to 54.8% with graph-temporal features added, and 59.0% with the full 448-feature set. Voltage level, relative angle, frequency, time scale and spatial coherence carry complementary information; Fig. 5(b) reports an executed feature comparison rather than an inferred explanation of tree importance. Electrical features make a classifier physics-informed in its representation. They do not enforce Kirchhoff balance on its predicted label or compute the PMU response of a candidate intervention.

For phase-to-neutral RMS phasors and a declared current orientation,

$$S_{ij} = P_{ij} + \mathrm{j}Q_{ij} = \sum_{\phi\in\{a,b,c\}} V_{i\phi} I^{*}_{ij,\phi}. \tag{17}$$

This expression requires voltage and current at the same terminal. In the frozen map, the bus-2 voltage is paired with a current measured at bus 3, and the bus-10 voltage with a current at bus 11. Their raw channel products are therefore power proxies, not exact terminal powers. Even a co-located terminal power is not the bus's net injection unless all relevant terminals are accounted for. Frequency and ROCOF complement the phasors but share the instrument's angle processing; the likelihood must account for that dependence. Fundamental-frequency phasors also do not determine harmonic distortion. Thus adding nominally physical features does not by itself supply missing harmonic or nodal-injection measurements.

C. MIXED-SOURCE RESPONSE COEFFICIENTS

For simultaneous load changes, a second-order response approximation includes products of the input amplitudes. In the archived two-load experiment,

$$s_{q,n}(a) = \sum_i D_{qni} a_i + \sum_i Q_{qni} a_i^2 + \sum_{i<j} Q_{qnij} a_i a_j, \tag{18}$$

where q indexes PMU coordinates, n elapsed reports and i,j load sources. Centered nonlinear simulations estimate the coefficients; the Q terms include their Taylor scaling. The mixed term represents the interaction of two simultaneous load changes. Across 32 archived two-source responses, median whitened error is 0.205 for D, 0.469 after self-curvature, and 0.00307 after mixed terms. This is evidence for response fidelity in that local experiment. It is not a measured classification gain from inserting this second-order tensor into the continuous load/line bank, which uses switched local predictors instead.

The coefficients in (18) form a tensor over measured coordinate, elapsed time and source indices. The useful term is the physical cross-source interaction Q_{qnij}; the array's shape alone does not provide an accuracy or computational advantage. In the continuous estimator, M is instead a vector of aggregated source/family weights at each report, and ΔM is its temporal difference, as defined in (15)–(16). That implementation does not require a tensor decomposition or a tensor-learning library.

D. WHY INTEGRATE SEVERITY RATHER THAN SELECT THE BEST FIT?

An extra source with zero amplitude can reproduce a smaller source set. Best-fit residual alone therefore cannot decide cardinality. In the archived load-support estimator, the model-selection quantity is

$$Z_S = \int p(r \mid s_S(a), \Sigma) p(a \mid S)\, da, \qquad p(S \mid r) \propto Z_S p(S). \tag{19}$$

Integration accounts for the prior volume of amplitudes consistent with the data, rather than rewarding only the best-fitting amplitude. This is standard Bayesian model comparison applied to an electrical response model, not a new Bayes rule. The archived implementation evaluates the null, 16 single-source and 120 pair supports with 31-point Gauss–Hermite quadrature per amplitude. Table 3 shows both its successes and its loss of resolution for weak sources.

An audit found a one-source evidence normalization offset in this archive: the implemented cardinality prior is pro-

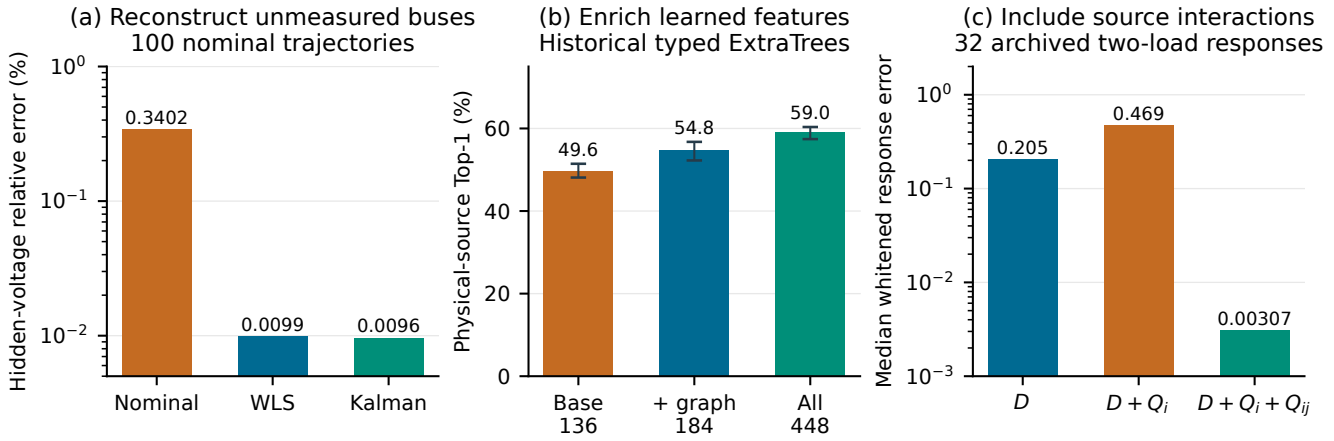


FIGURE 5. **Three separate component experiments. (a) Mean relative error of the 31 unmeasured bus voltages under a matched nominal model; this is not a PMU compliance TVE test. (b) Mean physical-source Top-1 over three model seeds, each scored on 7,915 labeled physical samples, in the historical typed-ExtraTrees feature study; whiskers show the seed range, not a confidence interval. These 136–448-feature models are neither the final hackathon submission nor the causal comparators. (c) Median whitened prediction error at the archived 120-report horizon for two-load responses: mixed-source terms matter, while adding only self-curvature worsens this metric. Each panel has its own units and cohort.**

TABLE 3. **Archived joint cardinality-and-support decisions. Onset, load family and operating context are supplied. These are noise replays of shared physical trajectories, not independent continuous detections.**

Amplitude regime	Replays	Joint support Top-1
Finite / finite	1,440	100.00%
Moderate / moderate	1,440	98.82%
Weak / strong	1,440	66.25%
Weak / weak	5,760	24.38%

portional to $(0.20, 0.50/(2\pi), 0.30)$ rather than the declared $(0.20, 0.50, 0.30)$. The table preserves the executed results under that effective prior; it does not validate calibrated posterior probabilities or a corrected replay. The continuous estimator uses separate, independently checked local evidence code. Consequently these component results motivate the physical/Bayesian construction, but do not establish that adding Bayes or tensor storage alone causes its final accuracy advantage.

VI. EXPERIMENTAL DESIGN

The test asks whether one method can attribute six successive load or line interventions in a continuous trajectory, including an intervention that begins while the preceding response is still recovering. The plant is a fixed Julia/PowerDynamics IEEE 39-bus research candidate. Each parent is integrated once across all ON and OFF actions; differential states are not reset. PMU reports are exported at 30 frames/s, independently of solver steps. A switch-aligned report uses the post-switch algebraic value. RAW0001 informed the simulator retrospectively and is not an independent test record.

The sealed evaluation has 18 six-event continuous parents, 27 persistent line-opening parents and 12 normal parents. Every parent is replayed under matched R/X, an R/X change represented in the model bank and an R/X change outside it. Two instrumental-noise scales give 171 physical trajectories and 342 observed records. The 648 continuous event realizations repeat 108 interventions; the physical parent, rather than the PMU frame or replay, is the uncertainty unit. The earlier development examples in Fig. 8 are not part of this sealed comparison.

A. WHAT THE FEATURE BASELINE ACTUALLY DOES

The hackathon submission was a hierarchical detector, event classifier and typed localizer based on ExtraTrees. It summarized each 30-s PMU window with voltage/current magnitude and angle statistics, frequency, ROCOF, missingness, phase spread, approximate P/Q, rolling changes, RLS/Kalman level innovations and graph-temporal differences. Its selected bundle reports 45,162 input features and internal simulated-validation macro-F1 0.960 with exact localization 0.852. These numbers describe that archive's broader nine-label, window-level task; they are not scores on the present event-level cohort. The original submission is preserved.

To compare at 30 reports/s, an earlier development campaign built two *new*, frozen causal ExtraTrees adapters from 47 PMU coordinates. Each coordinate is centered and robustly scaled using the quiet prefix. For trailing windows of 1, 5 and 10 s, the reference adapter computes median, standard deviation, 10th/90th percentiles, a recent-quarter change and RMS first difference. This gives $47 \times 3 \times 6 = 846$ features per report. The short-window adapter adds the current value and mean, standard deviation and two lag changes at 3, 8 and 15 reports, giving 1,457 features in total. Both use 400 trees, balanced class weights, 19 classes (quiet plus 18 assets), and six-report confirmation; tree votes are scores, not calibrated event probabilities. Their training parents precede and are disjoint from the sealed cohort. Neither model was retrained or threshold-adjusted on these outcomes. The original split contains 32 local physical IDs for fitting (64 records, 78,784 training rows) and eight for validation (16 records). A later dependency audit maps these to 26 fitting and eight validation

groups, with two normal-control groups shared across those sets. This limits independence of the internal validation; it is not evidence of leakage into the separate confirmatory cohort. The reference and short-window classifiers already obtain only 0/72 and 4/72 timely load attributions in that development validation. Their frozen performance therefore does not represent a successfully optimized load classifier.

A later grouped retraining study uses 31 fit, eight validation and 17 reserved physical groups: approximately 70/30 development/reserved, with an inner validation partition. That model is a separate artifact, not either comparator executed in the confirmatory study. Randomly splitting PMU reports 70/30 would allow correlated trajectories to cross partitions. RAW0001 is a known retrospective transfer check, not the reserved 30% or an external test. The supplement distinguishes these three evaluation contracts.

The sealed methods are the five-model physical bank, its nominal-network version and these two causal ExtraTrees adapters. All receive the same 30-frame/s PMU stream and retain their own frozen decision rules. This tests complete procedures at their declared operating points, not a matched-false-alarm or unlimited-supervision comparison. The physical banks also have different publication rules, so their contrast does not isolate network-model averaging.

Chen *et al.*'s published quickest-change detector [2] is eligible only for persistent line openings and normal records. The declared adapter uses nonoverlapping 15-Hz angle increments, a nominal single-line catalog and prefix-estimated drift and scale. It stops at its first alarm. Its DC and white-injection assumptions do not fully describe the simulator's colored PMU noise; no published average-run-length guarantee is transferred to these records.

For continuous streams, an alarm is matched temporally before inspecting its source. A timely decision requires the right family and asset before true OFF. Unmatched event-stream alarms are extras. Restoration messages on unmatched episodes are counted separately; this count is not every possible physical OFF-timing error. For persistent lines, the first stop is scored as correct, premature, wrong-source or censored. Failed records remain in the intended denominators. Paired 95% intervals resample physical parents 10,000 times. Logical delay uses report timestamps; wall-clock inference cost is a separate quantity. Full calendars, thresholds and reproduction details are in the supplement.

VII. DEVELOPMENT COMPARISONS AND WORKED SEQUENCE

A. DIRECT TRANSFER OF THE SUBMITTED HACKATHON SYSTEM

Before the confirmatory continuous study, the unchanged submitted ZIP and an earlier physical estimator were executed on the same 124 windows: 96 event realizations and 28 normal windows. Figure 6 preserves their actual family confusions. The submitted ExtraTrees pipeline detected 32 of the 96 events, assigned the right family to 23, and the right family and exact source to 0. The earlier physical estimator obtained 96/96 for all three quantities. Neither alarmed in the 28 normal windows. The source result was checked against the submission's output CSVs, not inferred from the family matrix.

This comparison exposes a transfer failure of the delivered system on the later simulator's load/line catalog. It does not show an official hackathon-score improvement. The submitted model retains its broader training domain and 30-s window rule, whereas the physical estimator knows the simulator and its 12 candidate origins. These were development cases, and the comparison permits both methods the same window endpoint. The next study therefore uses separately trained causal ExtraTrees comparators, successive events and a new frozen cohort. Its results must be read as a second experiment, not as more independent repetitions of the perfect development score.

B. FOLLOWING SUCCESSIVE EVENTS THROUGH RECOVERY

The example in Fig. 8 follows six physical interventions without restarting the network. With matched R/X, all six sources are identified before their OFF times. Under a network change outside the bank, an extra bus-20 diagnosis precedes the real bus-31 load; bus 31 is then diagnosed only after its physical restoration. The lower panel shows the mechanism visible to the method: active bus-31 mass rises late in the mismatched record. This is one development trajectory, not a performance estimate.

The development controls also show that deleting alarms is easy if true events may be lost (Fig. 7). The stricter onset witness has no extras but loses a real bus-16 intervention. The frozen five-model procedure was therefore evaluated with its original publication rule; no rule was revised from the sealed outcomes.

C. COMPARISON WITH A SIMPLER PHYSICAL PROCEDURE

An earlier development control uses the same nominal physical and PMU operators, one retained state and a short generalized likelihood-ratio search. With a common six-report publication clock, it obtains 37/54 timely decisions and 37 extras; the history mixture obtains 51/54 with 3. This is an executed physical comparator, separate from the later confirmatory cohort. Different search, severity treatment and score semantics prevent it from being a pure Bayes or memory ablation. Its equations and full table are included in the supplement.

VIII. SEALED EVALUATION: ACCURACY AND ALARM COST

The five-model bank makes 644/648 timely family-and-source decisions with 72 unmatched event-stream alarms. Its nominal counterpart makes 643/648 with 92 extras. The frozen causal ExtraTrees reference makes 330/648 with 29; its short-window variant makes 352/648 with 16 extras. These are paired replays of 108 interventions, not 648 independent events. Figure 9 shows why the totals differ: the physical

TABLE 4. **Three distinct inference contracts. Historical hackathon validation and the sealed causal comparison have different units and cannot be subtracted as if they were the same score.**

	Hackathon submission (historical)	Causal ExtraTrees comparators	Physical history bank
Input and clock	Eight PMUs; one decision per 30-s window	Same 47 PMU coordinates as physical bank; 30 reports/s	Same 47 coordinates; 30 reports/s
Information used	Engineered window features; supervised family/location labels	Trailing statistics; supervised active-asset labels from earlier simulated parents	Nominal DAE, candidate asset operators, PMU terminal map and quiet prefix
Decision	Hierarchical detector/classifier/localizer	Tree vote, then six-report episode confirmation	Conditional history evidence, active/cause masses and fixed publication rules
Tested scope here	Not rerun on the sealed cohort	Nine loads, nine lines, successive events	Same nine loads and nine lines, successive events
Main limitation	Window-level historical score is not a causal event-level score	May miss load sources absent from its learned feature boundaries	More extra alarms under out-of-bank R/X

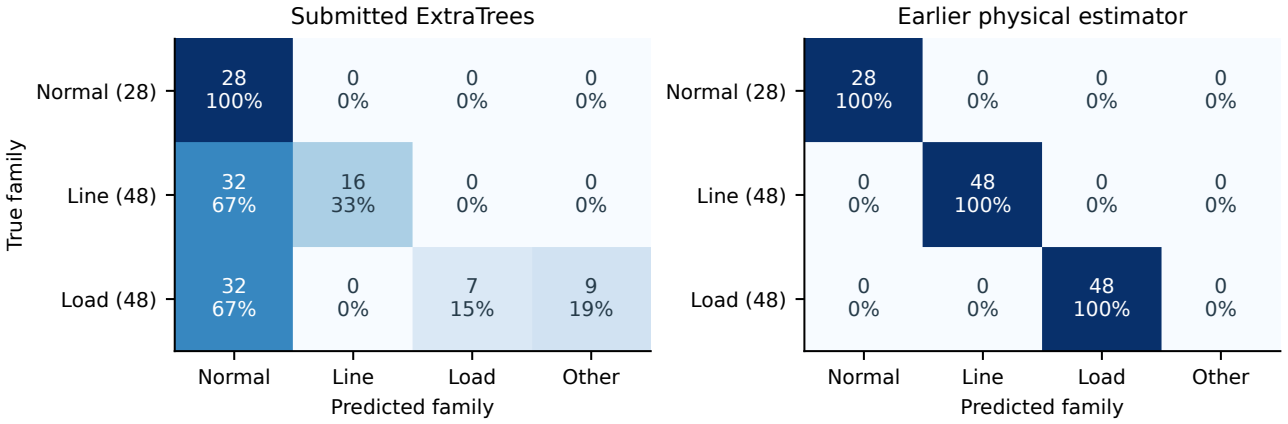


FIGURE 6. **Historical development transfer on shared observation windows. Cells show counts and row percentages. All out-of-scope predictions are retained as "Other"; the submitted system's nine such predictions were the unknown-event label. This is a family matrix, not a localization matrix. The physical comparator is an earlier estimator, not the final continuous bank. The 96 event realizations reuse 36 physical event trajectories; the normal windows come from six physical trajectories.**

TABLE 5. **Sealed continuous-stream decisions. Each family has 324 replayed instances from 54 distinct interventions (108 across both families); extras are unmatched alarms on event records. The methods retain unequal alarm operating points.**

Method	Load	Line	Extras
Physical bank	322/324	322/324	72
Nominal bank	322/324	321/324	92
ExtraTrees	15/324	315/324	29
ExtraTrees short	30/324	322/324	16

banks recover nearly all loads, while both learned adapters already recover most tested lines. The extra alarm burden of the physical banks rises sharply when true R/X changes fall outside their five-model bank.

A posthoc audit of the original episode matches further separates coverage from classification. Every matched diagnosis emitted before OFF has the correct family and source in all four procedures; their errors in the timely endpoint are missing or late diagnoses. The load advantage therefore concerns timely coverage. It does not demonstrate higher conditional classification accuracy among diagnoses already available before OFF. The supplement shows the complete confusion view and counts unmatched alarms separately.

The paired intervals in Fig. 10 confirm a timely-attribution advantage over this ExtraTrees reference in each tested stratum. They also show a higher rate of unmatched alarms under out-of-bank R/X. Those intervals resample physical parents, preserving correlation across noise replays and network conditions. They do not establish superiority at a common false-alarm operating point. The physical bank's stratum median diagnosis delay is about 0.27 s in matched and in-bank conditions; the ExtraTrees medians range from 0.19 to 0.30 s. There is no general latency advantage.

The separate persistent-line test scores the first stop. Both physical banks and both ExtraTrees adapters identify the right line first on 162/162 openings. The declared Chen2016 adapter does so on 74/162; 66 records stop before the opening and 22 stop afterward on a wrong line. A separate descriptive audit of the same persistent records shows 164 and 179 later unmatched alarms for the two ExtraTrees procedures, versus 0 for the physical bank. Correct first identification therefore

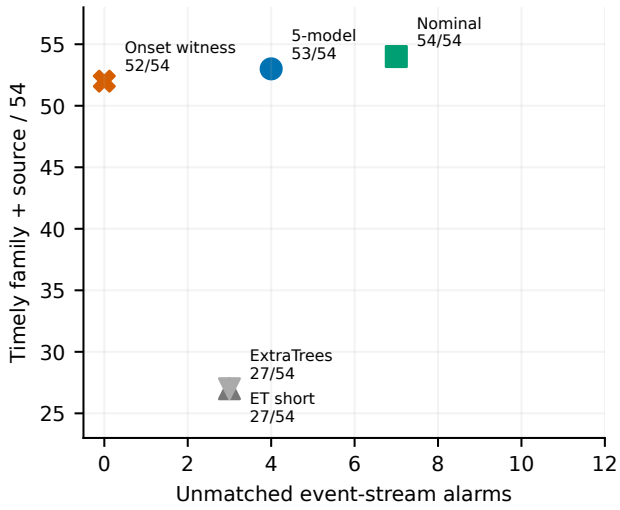


FIGURE 7. **Development decision trade-off on 18 interventions replayed under three R/X settings. A stricter onset witness eliminates extra alarms but misses a true bus-16 load. The 54 plotted realizations share physical parents. These controls were not selected using the later sealed evaluation.**

does not imply stable maintenance of a sustained event. This audit does not change the first-stop comparison or apply a recurrent score to Chen. Chen also stops on all 72 paired normal records, whereas the continuous methods alarm on none. Chen's normal exposure ends at its first stop (2,522 s); the other methods monitor 42,336 s each. These unequal exposures and the adapter's white-injection assumption prevent a general claim against published quickest-change detection. Full stratified tables, failure ledgers, restoration outcomes and censoring are in the supplement.

The experiment establishes performance on the specified nine-load, nine-line simulator catalog. It does not establish a new hackathon score, unseen-source generalization, calibrated event probabilities, field operation or real-time throughput.

IX. EPISODE RESTORATION AND COMPUTATIONAL COST

Restoration timing gives a second view of continuous attribution. Figure 11 shows the signed OFF delay among available associated restorations, alongside intended events without a restoration value. Under the posthoc threshold of more than one 30-Hz report early ($d_{OFF} < -1/30$ s), the physical bank has 0/644 early restorations and the nominal bank 0/643. The reference and short-window ExtraTrees have 52/330 and 26/352, respectively. The corresponding missing-restoration counts are 4, 5, 318 and 296 out of 648 intended event realizations. Missing values include omissions and are not necessarily open episodes. Replays are dependent; this descriptive criterion is not a new confirmatory endpoint and does not isolate the memory component.

Recorded stage timings from the continuous evaluation are summarized in Table 7. For 120-s records, median physical-history inference took 358.7 s across 270 records, compared

TABLE 6. **Posthoc description of episode restoration in the continuous evaluation. Early counts restorations with $d_{OFF} < -1/30$ s (more than one report early); Early (%) divides by Available. Missing counts absent restoration values out of 648 intended events, including omissions and not necessarily open episodes. Replays are dependent; no significance test is implied.**

Method	Available	Early	Missing	Early (%)
Physical bank	644	0	4	0.0%
Nominal bank	643	0	5	0.0%
ExtraTrees reference	330	52	318	15.8%
ExtraTrees short	352	26	296	7.4%

TABLE 7. **Recorded stage times from the continuous evaluation on 120-s records. Medians and ranges summarize full-record stages; the per-report column is a median-total divided by 3,601 reports and is amortized, not frame latency. Stages were recorded in a campaign with one BLAS and one Julia thread; memory and per-report latency were not recorded. This descriptive table does not establish real-time operation or a speedup under matched implementation conditions.**

Stage	Median (s)	Range (s)	ms/report†
Bank build	80.1	62.0–309.7	22.2
History inference	358.7	218.2–2641.1	99.6
ExtraTrees (both)	9.5	6.7–50.5	2.6
Chen detector	1.3	0.9–10.6	0.4

with 9.5 s for the combined ExtraTrees adapter stage; physical bank construction had a separate 80.1-s median. Dividing total time by 3,601 reports gives amortized averages of 99.6 ms/report for physical inference and 2.6 ms/report for the combined learned stage. These totals include full-record work and do not measure per-report latency. The physical stage is slower in this recorded implementation and does not support a real-time claim. Timers span mixed stages from the evaluation and concurrency; memory use and frame-level latency were not recorded. The 600-s normal controls are summarized separately in the machine-readable cost file.

X. CONTROLLED COMPONENT STUDY

The continuous comparison evaluates complete procedures. To distinguish two mechanisms within the physical bank, we fixed a subsequent 2×2 study: marginal amplitude evidence versus a bounded profile score, and physical-state carry versus removal at inferred closure. The five network models, observations, startup fits, operators, proposal rules and publication thresholds are common to all four arms. Identical thresholds do not imply identical false-alarm rates; both coverage and alarm counts must be measured.

For one load history and amplitude interval $[l, u]$, write its accumulated conditional log likelihood as

$$\ell_k(a) = c_k + b_k a - \tfrac{1}{2} G_k a^2. \tag{20}$$

The original score uses $B_k = \log \int_l^u \exp(\ell_k(a) - c_k) p(a)\, da$ with the existing truncated Gaussian prior. The alternative uses

$$\widehat{a}_k = \mathrm{clip}_{[l,u]}(b_k/G_k), \qquad Q_k = b_k \widehat{a}_k - \tfrac{1}{2} G_k \widehat{a}_k^2, \tag{21}$$

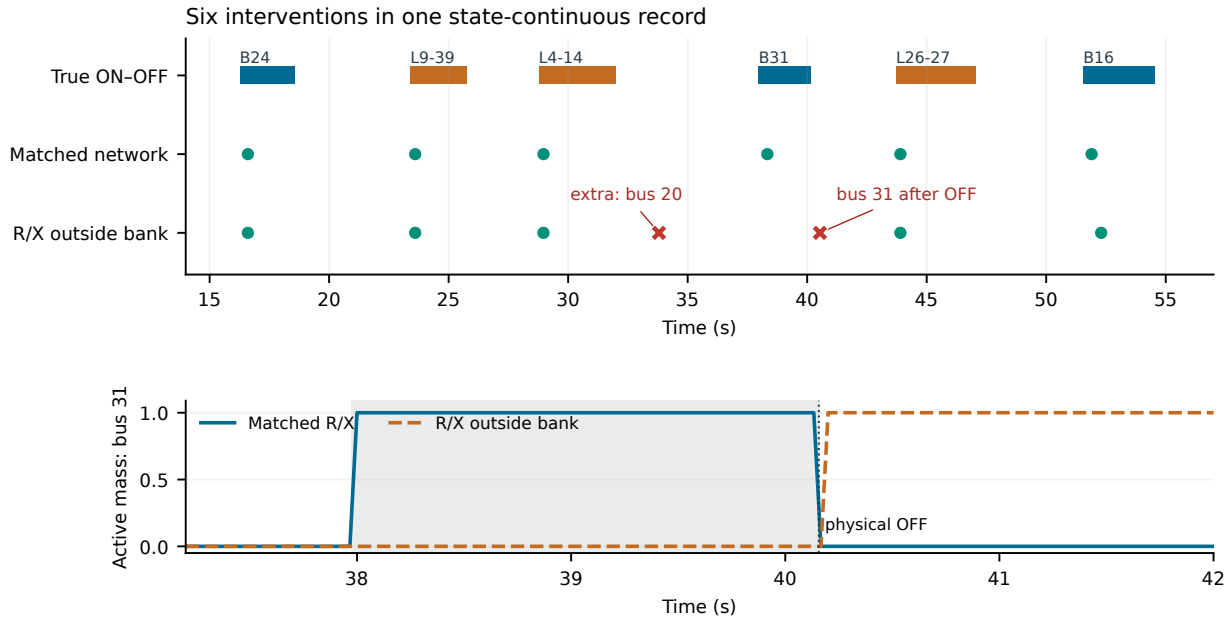


FIGURE 8. **One development sequence. Bars show reserved physical ON–OFF truth; circles are correct diagnoses emitted before OFF; crosses mark an extra bus-20 alarm and a bus-31 diagnosis emitted after OFF. Both lower decision rows use the same physical bank under different R/X conditions. The lower panel shows approximate active mass for bus 31: with R/X outside the bank, the correct explanation becomes dominant only after restoration. Truth is shown retrospectively and never enters inference.**

with the constant-likelihood case treated separately. Each report adds $B_k - B_{k-1}$ or $Q_k - Q_{k-1}$, so a rolling replay does not repeatedly add the same suffix evidence. The conditional Gaussian amplitude/state updates remain unchanged. Profile weights are normalized scores, not posterior probabilities. This intervention tests the amplitude-scoring choice, rather than removing every Bayesian calculation. The existing Gaussian sufficient statistics supply $G_k = v_k^{-1} - v_0^{-1}$ and $b_k = \mu_k / v_k$, removing the computational prior precision before profiling. Fixed line-opening amplitudes and the prior weights of amplitude intervals are unchanged.

At an internally inferred closure, the carry arm retains the projected augmented state. The reset arm sets its 114 estimated physical-state deviations to zero and retains instrumental/discrepancy coordinates, covariance regime and network-model weights. This is an estimator ablation: every observed record comes from the same continuous physical trajectory. It does not test removal of all temporal information.

The protocol uses all 18 matched-network continuous parents at the original noise level, plus the first three normal and persistent-line parents by identifier. It was fixed before ablation outcomes, but the underlying records had already been examined. Results are therefore a posthoc mechanism study, with paired intervals over physical parents, not a new independent confirmation.

A. COMPONENT OUTCOMES

Table 8 and Fig. 12 report coverage and alarm cost for the four arms. With carry retained, marginal and bounded-profile amplitude scores tie on the reported endpoints. Resetting the first 114 estimated physical-state deviations at inferred closure reduces timely attribution and increases extras under both scores. This reset leaves instrumental state, covariance regime, prior weights, thresholds, clocks and reporting rules intact; simulated states and observations never reset. The profile arm remains a normalized penalized score system and retains conditional Bayesian state estimation. Thus this study supports a physical-state-carry effect at inferred closure in the tested cohort, not the necessity of Bayes or of every form of temporal memory. The supplement reports denominators, normal exposure, persistent episodes, censoring and all paired parent-cluster intervals. These are posthoc within-simulator contrasts at the original thresholds, not a comparison at matched false-alarm rates.

XI. INTERPRETATION AND LIMITS

A. PREDICTION IS NOT IDENTIFICATION

For the exact observation map Ψ, compare a new intervention e with and without the same earlier history h:

$$s_{e|h}(t) = \Psi_t(h \oplus e) - \Psi_t(h). \tag{22}$$

Both paths start from the same physical and instrumental state. For an additive intervention under the same linear time-invariant dynamics, this increment is independent of the prior state. A topology change can alter the state matrix, so that statement does not apply to switched linear models in general. Neither switching nor nonlinearity guarantees more source information. Our stream implementation approximates both paths with local switched predictors. It does not solve the full nonlinear DAE for every history.

A good fit to one source does not establish that competing sources can be separated. With severity and nuisance coordi-

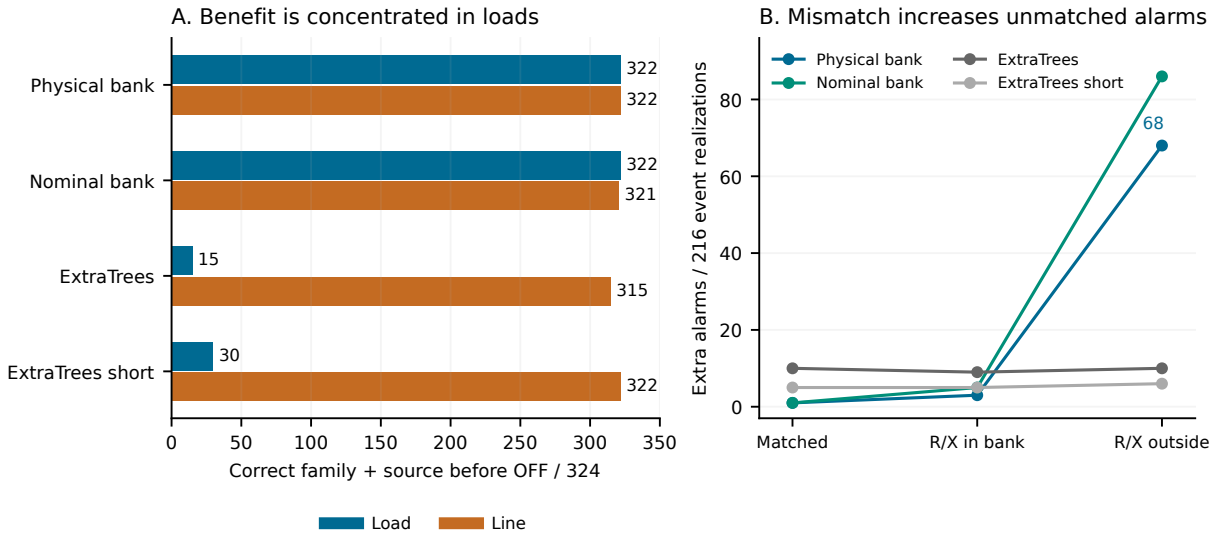


FIGURE 9. **Sealed continuous evaluation. (A) Timely correct family and source by event family, with 324 paired realizations per family. (B) Unmatched event-stream alarms across the two noise levels within each R/X condition, with 216 event realizations per condition. The plots display the gain in load attribution and the alarm cost under out-of-bank network error together. Each method retains its own frozen publication rule and false-alarm budget.**

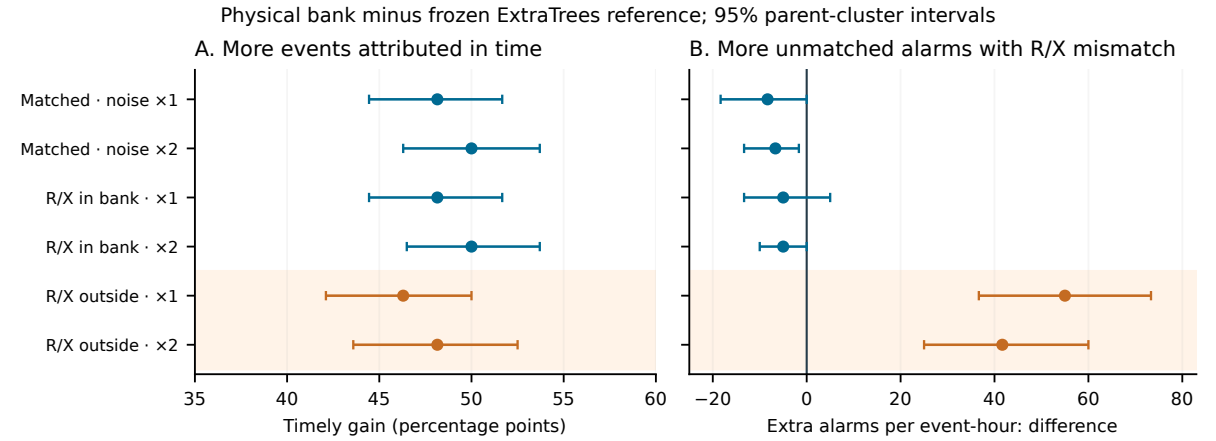


FIGURE 10. **Physical bank minus frozen ExtraTrees reference in the six paired strata. Points and 95% intervals come from 10,000 physical-parent resamples. Left: percentage-point difference in timely attribution; right: difference in unmatched alarms per event-record hour, whose denominator includes the 12-s startup. The orange rows expose the out-of-bank R/X failure. The intervals describe this finite simulated population.**

nates β_j profiled over their admitted set $\mathcal{A}_j$, a local separation measure is

$$d_h^2 = \min_{j \neq h} \min_{\beta_j \in \mathcal{A}_j} \|s_h - s_j(\beta_j)\|^2_{\Sigma^{-1}}. \tag{23}$$

Small distance makes location noise-sensitive; it is neither a universal error bound nor a calibrated posterior [9], [10]. Fit, source ranking and uncertainty calibration require separate tests. The supplementary controls retain their known-onset and observation contracts; their state reconstruction scores are not pooled with the causal study. Severity fitting can absorb context error parallel to a source signature while leaving intensity biased. This standard projection effect explains why correct location and inaccurate response amplitude can coexist.

B. WHAT RECOVERY MEMORY ESTABLISHES

A separate full-DAE control supplies the earlier history and later onset. Source separation stays essentially unchanged when that history is known and falls when prior severity is uncertain; binary source accuracy has no mean gain. Retaining recovery is therefore justified as a consistent prediction of the continuing response, not as a universal increase in location information. The stream comparison tests the complete frozen procedure. Its admission-rule ablation does not isolate every operation that retains recovery.

A separate signature test holds pre-event measurements and offset equal while changing the event–context response term. It isolates a response approximation, not an improvement of the unchanged causal detector. Known timing, one scalar context and four load sources bound that result.

TABLE 8. Controlled amplitude-score and physical-state-carry ablation. Same five-model bank and publication rules. All 18 continuous parents (108 events) are retained. Normal controls comprise three 600-s records with 588 s evaluated exposure each (1,764 s total; startup excluded). Delay is conditional on emitted matched diagnoses and must be read with the no-timely-diagnosis count.

Procedure	Timely	Extras	Normal FA	False restorations	No timely	Delay (s)
Marginal + carry	108/108	1	0/1,764 s	1	0/108	0.267
Profile + carry	108/108	1	0/1,764 s	1	0/108	0.267
Marginal + reset	75/108	166	0/1,764 s	130	33/108	0.337
Profile + reset	75/108	166	0/1,764 s	130	33/108	0.337

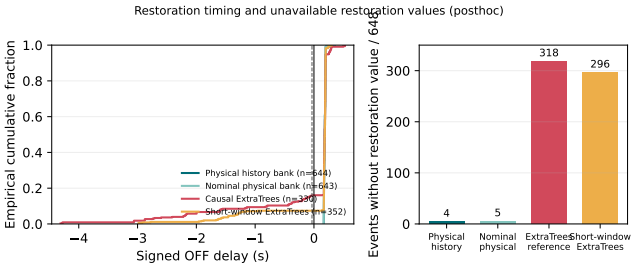


FIGURE 11. Posthoc restoration timing and events without an associated restoration value. Panel A shows signed OFF delay only where available; the dashed line marks the threshold at one report early. Panel B reports missing restoration values over 648 intended events, including omissions, not a count of episodes open at record end.

C. WHAT THE COMPARISON DOES NOT ESTABLISH

Trajectory libraries, sequential likelihoods and multiple-model filters precede this work. The claim here is the specified competition of new interventions with retained recovery, and its measured behavior, rather than a general filtering optimum. Warner's trajectory procedure was not run on this cohort. The feature baseline's trailing summaries describe how signals changed, but do not encode the counterfactual response of *each candidate source* through the network and PMU terminal map. That is a concrete difference in information used by the algorithms and is consistent with the observed load gap; these experiments do not prove that a better feature set or larger training set could not close it. The original hackathon submission addressed nine labels with 30-s decisions. It is a useful earlier result and a source of feature design, but its internal validation score cannot be compared numerically with event-level timely attribution here. The two causal ExtraTrees adapters were trained separately for this task; their load misses do not establish a new official hackathon score or rule out better supervised models. Chen's first-stop adapter is tested only on persistent lines under DC and white-injection assumptions.

D. BROADER HACKATHON SCOPE AND RETROSPECTIVE TRANSFER

The earlier all-family pipeline and integrity interface are retained as separate results. An archived observation-only composition reproduces all 180 physical/integrity windows and reconstructs availability over 161379 reports; this verifies interface consistency, not additional detection accuracy. On retrospectively annotated RAW0001 windows, a source-only learned hybrid changes joint family/source correctness from 1/24 to 2/24, with one gain and no losses. It is not the physical bank and does not establish broad transfer. The frozen physical bank rejects that record because the prefix ROCOF noise scale is outside its admitted support; no later quiet prefix was substituted to force admission. The supplement preserves these outcomes alongside the simulation results.

E. SIMULATION AND DEPLOYMENT BOUNDARY

The causal catalog has nine load buses and nine lines, with a quiet startup. Missing PMUs, unknown families, unrestricted simultaneous sources, unseen-source transfer and field records remain untested. The generating plant and inference predictors share the F01 equation family and PMU map. The three R/X conditions perturb parameters within that family; they do not test a different producer, control law or instrument installation. Thus the high within-simulator hit rate is not evidence of the same rate on RAW0001 or a field network. The simulator keeps a provisional BUS5 current terminal and omits BUS39 current angle. Its noise includes temporal and cross-PMU dependence within signal families but omits the archived cross-family factor; frequency/ROCOF noise is added after phase-based estimation as separate signal families. It is not obtained by differentiating the exported noisy angle, and the archived cross-family noise factor is not applied. Hence agreement with this simulator does not verify the source producer's CT placement or a complete physical PMU installation.

Out-of-bank R/X preserves most timely source decisions but sharply raises unmatched alarms. The parent-cluster intervals describe these simulated parents, not all grids or PMU layouts. Context fitting, history pruning and shared covariance approximations also prevent a calibrated-posterior claim. Report-causal operation has not been timed as a real-time service; the implementation is not validated for protection or control.

XII. CONCLUSION

The proposed procedure attributes load and line interventions by comparing their predicted PMU responses with the observed stream. Each candidate starts from an estimated past state. Restoration changes its input or topology while its physical state continues to evolve. The method integrates local load severity and reports event activity separately from the cause of a continuing response.

The sealed continuous evaluation gives 644/648 timely

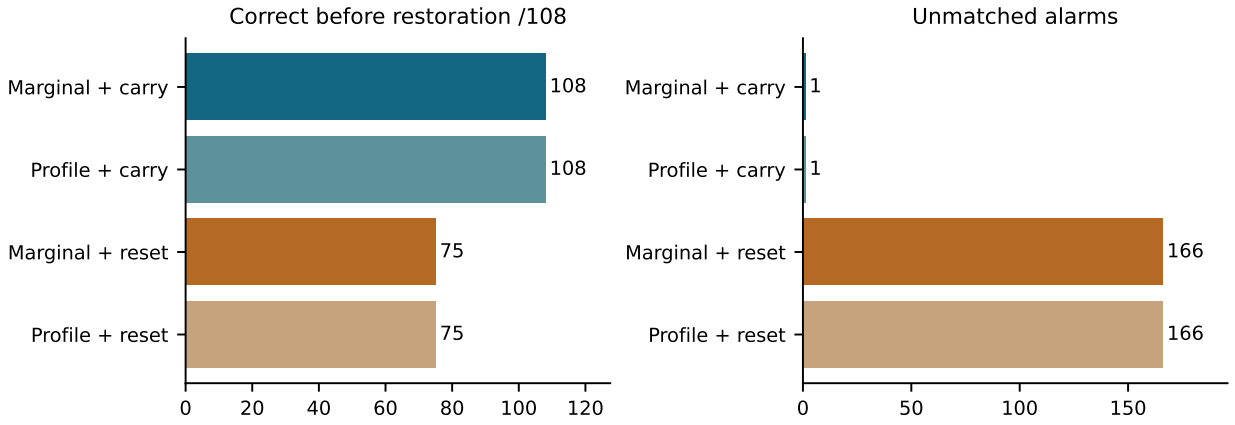


FIGURE 12. **Amplitude-scoring and physical-state-carry ablation on identical observations and five-model banks. The 108 interventions belong to 18 continuous trajectories. Thresholds are fixed, so the observed alarm rate may differ between arms.**

family/source decisions with 72 extra alarms for the mixture, and 643/648 with 92 for nominal. The frozen ExtraTrees reference gives 330/648 with 29 extras. Most mixture extras arise under out-of-bank R/X, so the gain in timely attribution does not imply uniformly better alarm control.

The component experiments preserve useful earlier findings: network conditioning recovers hidden voltage information, graph-temporal features improve a historical learned localizer, and mixed-source terms reduce local response error. They do not establish an isolated causal gain from Bayes or tensor storage in the final estimator.

The posthoc component study sharpens the recovery result. On the same 18 continuous trajectories, both score arms with physical-state carry attribute 108/108 events on time with one extra alarm; both reset arms attribute 75/108 with 166 extras. Marginal and bounded-profile scores tie on these endpoints, while the profile arm retains conditional Bayesian state estimation. The observed contrast supports physical-state carry at inferred closure in this cohort. It does not establish the necessity or dispensability of Bayesian state estimation, prior weighting, or other temporal information.

The earlier hackathon pipeline remains a broader nine-label, feature-based system. Its 30-s-window validation is a separate result; the causal ExtraTrees successor in this paper provides the direct same-stream comparison. The physical bank's strongest supported gain is timely load-source attribution, while learned features already diagnose most lines in this catalog with fewer unmatched alarms.

The evidence concerns a specified load/line catalog in one simulator. Unmodeled R/X increases the alarm burden. The controlled component study addresses amplitude scoring and state carry at inferred closure; it does not remove all Bayesian estimation or all temporal memory. All-family diagnosis, calibrated event probabilities, field transfer and real-time deployment remain outside the demonstrated scope.

A source-excluded comparison would test whether the procedure generalizes beyond its supported catalog. A learned correction of physical residuals could address model mismatch, provided it is evaluated independently of source-label fitting. Joint state/context uncertainty and event-probability calibration also require further validation.

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